



Linux OS – Lab 3

Lab Description:

In this lab, you will explore the Linux operating system and gain hands-on experience with fundamental tasks. You will boot into the CentOS operating system, navigate its interfaces, and perform common operations using the terminal. This lab complements the theoretical concepts covered in the lecture.

Task 1: Linux File System (Revisited)

- Perform the following:
 - Navigate to **/var/tmp** using **cd** command.
 - Use **pwd** to confirm your current directory.
 - Use **tree** command to explore a directory structure.
 - Navigate to the **Downloads** directory using **cd** command.
 - Create the following directory structure using **mkdir** and **touch** commands then navigate it using **cd** and list its contents using the **ls** command:

```
project/
├── src/
│   ├── main.py
│   └── data.js
├── bin/
│   └── conf.ini
├── docs/
└── .gitignore
```

- Open the current working directory in the GUI using **xdg-open** command.

Task 2: Managing Files and Directories

- Create 5 files using the **touch** command named **file1.txt**, **file2.txt**, ..., **file5.txt** using **touch** command with file globbing. (**Hint**: {1..5} expands to 1 2 3 4 5)
- Copy **file1.txt** and **file2.txt** to a new directory called **backup/**.
- Display the contents of **file1.txt** and **file2.txt** using **cat**.
- Remove the **backup/** directory and confirm its deletion using **ls** command.



Task 3: File Searching and Patterns

– Using Wildcards

- Create **fileA.txt** and **fileB.txt**.
- List all files starting with "fil" and having a .txt extension.
- List all files ending with .t<t>

– Finding Files (Part 1)

- Use **find** command to find files starting with "F" using the find command and execute the file command on them.
- Perform the same task using the **locate** command. Discuss the differences between **find** and **locate** commands.

– Finding Files (Part 2)

- Use **find** to locate all files in **/home** modified in the last 7 days.
- Use **find** command to list all files larger than 5 MB in **/var** directory.
- Use **find** to find all directories containing **apache** in their name.
- Use **locate** to find files containing the term "conf".



Task 4: File Comparison

- Create the following files:

comp1.txt	comp2.txt
1234	4567
Mohamed Gamal	Ahmed Hassan
Cairo, Egypt	Cairo, Egypt
24 123 Main Street, Egypt	30 456 Secondary Street, Egypt

- Compare the two files using **diff**, **sdiff**, and **vimdiff** commands.



Task 5: File Sorting

- Create **users.csv** file with the following content:

users.csv					
1,	Mohamed,	30,	Egypt,	5000	
2,	Hassan,	17,	Egypt,	3000	
3,	Ahmed,	40,	USA,	10000	
4,	Rawan,	26,	UK,	7000	
Cairo, Egypt					

- Sort the file using the **sort** command in the following ways:
 - Alphabetically (ascending and descending).
 - By the second column, second and third columns, and numerically by the second column.
 - Remove duplicates.
- Use **wc** to count the number of lines, words, and characters.



Task 6: Text Search

- Create a **secret.txt** file with the following content:

secret.txt
site: youtube.com
user: Mohamed
pass: Abee123!
user: Ali
pass: Bcdd456!

- Use the **grep** command to:
 - Search for the word "user" (case-insensitive).
 - Search for the character "o".
 - Find lines that do not contain the letter "a".
 - Find the line number containing the word "user".
 - Count occurrences of the word "user".
 - Find all lines excluding the word "user".



Task 7: Using Pipes

- Execute **ls** and pass its output to **wc -l** using the pipe symbol (|) to count the number of files in your current directory.
- Pass the output of **\$ cat secret.txt** to the **grep** command to search for the word "pass".
- Use the **ps aux** command and **pipe (|)** its output to the **grep** command to find all processes related to **Python**.
- Use a command that **sorts** the output of **ls -lh** by file size in ascending order.

Task 8: Tab Completion and Backslashes:

- Use tab completion to execute a command with a long file name.
- Use backslashes (\) to execute a long command across multiple lines.